# M13D — Mock exam 4

Instruments · Rotary Wing Aerodynamics · Autoflight · Equipment and Furnishings

40 questions · 20 Instruments · 13 Autoflight · 5 Rotary Wing Aerodynamics · 2 Equipment and Furnishings

Level 3: 22 · Level 2: 13 · Level 1: 5. Three options, one correct.

Every question sits on a fact highlighted or underlined in the manual. No question in this paper appears in any of the other four.

The answer key follows the questions, with the explanation and the manual PDF page for each one.

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## INSTRUMENTS

1. An exhaust gas temperature indicator is fitted in a classical cockpit. Which group and subdivision does it belong to?
   - A. Aircraft monitoring, as an engine monitoring device
   - B. Flight monitoring, as a pressure monitoring device
   - C. Aircraft monitoring, as an airframe monitoring device

2. One screen in the cockpit carries all the air data information and also every Basic T parameter. This screen is
   - A. the HSI (Horizontal Situation Indicator)
   - B. the PFD (Primary Flight Display)
   - C. the standby horizon

3. The pitot probe of a simple system becomes completely blocked while the static system stays clear. Which of the instruments supplied by that system is affected?
   - A. The airspeed indicator
   - B. The altimeter
   - C. The vertical speed indicator

4. The Mach module of an analogue air data computer fails. Why is the static source error correction affected as well?
   - A. The correction depends primarily on the calibrated airspeed, which comes from that module
   - B. The correction depends on the static pressure, which is routed through the Mach module
   - C. The correction depends primarily on the Mach number, which comes from that module

5. The static pressure reaching an analogue air data computer is in error. What is the effect of this on the Mach indication
   - A. The Mach number stays correct, because the Mach module has a pressure input of its own
   - B. The Mach number is wrong as well, because the Mach module works from the other two modules
   - C. The Mach number is wrong as well, because the Mach module works from the temperature input

6. An aircraft climbs steadily and the MMO limit falls. Why does it fall?
   - A. Because the dynamic pressure decreases
   - B. Because the compressibility error increases
   - C. Because the speed of sound decreases

7. An aircraft makes only small changes of height. What happens to the coarse synchro value in the electrical altimeter?
   - A. It is blocked by the OFF flag monitor
   - B. It is blocked by the LIMITER
   - C. It is blocked by the HIGH LEVEL detector

8. How does the air data computer obtain true airspeed from equivalent airspeed?
   - A. It adds the static source error correction to the equivalent airspeed
   - B. It replaces the standard pressure used before with the real pressure
   - C. It replaces the standard density used before with the real density

9. The analogue output data of a hybrid air data computer is provided in the format
   - A. ARINC 629
   - B. ARINC 429
   - C. ARINC 565

10. A digital air data computer has lost its angle of attack, flap and gear position inputs. Which of its functions is degraded?
   - A. The corrections, such as the static source error correction
   - B. The conversion of the input pressures into electrical signals
   - C. The transmission of the computed values on the output busses

11. Air data modules are located near the static port or pitot probe, where they
   - A. amplify the air pressure for transmission along the tubes
   - B. convert the air pressure directly into a digital word
   - C. convert the air pressure into a proportional analogue voltage

12. Engineering wants an engine left on the wing as long as it is sound, yet changed at the home base before it causes trouble. Which installed system makes that possible?
   - A. The on board maintenance system, which stores reported faults
   - B. The flight data recording system, which records for the investigators
   - C. The monitoring system, which records data from various systems

13. An operator wants more than 25 hours of flight parameters held in the recorder. What makes that possible?
   - A. Using a shorter word than twelve bits
   - B. Using a second recorder in parallel with the first
   - C. Using the right combination of capacity and data rate

14. An aircraft is fitted with a glass cockpit. How is the rate of turn presented to its crew?
   - A. by the ball of the inclinometer
   - B. on an electronic display
   - C. on a pointer over a mechanical scale

15. A compass measures heading with a mounted, freely rotating permanent magnet. Which method is this and which kind of compass?
   - A. The mechanical method, a direct reading compass
   - B. The mechanical method, a remote reading compass
   - C. The electronic method, a remote reading compass

16. An aircraft is fitted with only the minimum number of display computers, so what does it lack compared with a large aircraft?
   - A. The computer that drives the centre displays
   - B. The computer that is there for redundancy
   - C. The computer that handles the analogue inputs

17. A pilot selects the ILS mode on his ND, so which two deviations does he then see?
   - A. Localizer and glideslope
   - B. Localizer and track
   - C. Glideslope and vertical speed

18. The take-off configuration check is run with a cabin door still open, so what will the crew find?
    A. The normal message will not appear
    B. The check will be bypassed automatically
    C. The master caution will be inhibited

19. Until when do the mode 3 visual annunciators remain active
    A. Until the landing gear has been retracted
    B. Until the warning envelope has been left
    C. Until a positive rate of climb is established

20. An engine vibration reading is seen in the cockpit, so how is its level expressed?
    A. Usually in millimetres
    B. Usually in units
    C. Usually in percent

## AUTOFLIGHT

21. What do the regulations demand for the operating states or modes of the autoflight subsystems?
    A. Feedback to the pilot by an aural warning tone
    B. Feedback to the pilot by coloured annunciators
    C. Feedback to the flight management system by a data bus

22. Which connection is actually made when an autopilot is engaged
    A. The torque limiter connecting the AP-servo with the trim actuator
    B. The servo valve connecting the AP-servo with the hydraulic supply
    C. The clutch connecting the AP-servo with the primary controls

23. A pilot holds a steady force on the control wheel in CWS. What do the inner loops produce from it, and what does the aircraft do?
    A. A proportional roll rate, which changes the attitude and the heading
    B. A fixed roll rate, which changes the heading but not the attitude
    C. A proportional roll angle, which changes the turn rate and the track

24. An upper pitch mode is switched off in flight on an aircraft whose basic pitch mode holds the momentary rate of climb. Which mode does the autopilot then adopt?
    A. Altitude hold mode
    B. Altitude acquisition mode
    C. Vertical speed mode

25. An aircraft is levelling off in altitude acquisition mode above the transition level. Which reference is the selected target expressed against?
    A. A radio height above ground
    B. A QNH corrected altitude
    C. A pressure altitude

26. The crew has not pressed the altitude hold button. Under what condition does the altitude hold mode still come in automatically?
    A. On approaching a pre-selected altitude with acquisition armed
    B. On approaching a pre-selected altitude with acquisition active
    C. On passing the transition altitude during a climb or descent

**27.** The aircraft is being steered along the centre line after landing — which two signals does the yaw controller process?

- **A.** The heading error and the glide path deviation
- **B.** The crab angle and the localizer deviation
- **C.** The heading error and the localizer deviation

**28.** An aircraft is flying on a path whose angle equals its potential flight path angle — what are its errors?

- **A.** Airspeed error zero and flight path error zero
- **B.** Airspeed error at a maximum and flight path error at a maximum
- **C.** Airspeed error zero and flight path error not zero

**29.** The aircraft accelerates in the cruise — what happens to the yaw rate signal passed on to the actuator?

- **A.** It is cut back as the speed rises
- **B.** It is built up as the speed rises
- **C.** It is held at its take-off value

**30.** The fail passive automatic flight control system fails during an approach. What must happen then?

- **A.** the system continues the approach and landing by itself
- **B.** the pilot has to take over the control of the aircraft
- **C.** the second autopilot takes over control and lands

**31.** The autopilot of a hybrid landing installation fails on short final. How can the landing still be completed?

- **A.** manually, following the supporting guidance system
- **B.** automatically, by the second autopilot taking over
- **C.** automatically, after the voter has selected a channel

**32.** The cyclic stick is moved to the left-hand side. Which servo stays unenergized and which movement results?

- **A.** the front servo, and the helicopter moves nose down
- **B.** the LH servo, and the helicopter moves sideways to the left
- **C.** the front servo, and the helicopter moves sideways to the left

**33.** The SCAS has received a rate signal from one of the gyros. What does it drive with its attenuation signal?

- **A.** the trim actuators of the cyclic stick
- **B.** the servo valves of the control servos
- **C.** the main servo input levers directly

## ROTARY WING AERODYNAMICS

**34.** Which force produced by the main rotor blades holds the helicopter up against gravity?

- **A.** Thrust
- **B.** Lift
- **C.** Drag

**35.** A helicopter is in forward flight and one blade is on the advancing side. In which direction does the relative wind on that blade act?

- **A.** Horizontally backward, opposite to the blade movement
- **B.** Horizontally forward, with the blade movement
- **C.** Backward and upward, across the blade movement

**36.** What must happen to the plane of rotation when an external force tilts the axis of rotation

- **A.** It tilts with the axis, staying perpendicular to it
- **B.** It tilts the opposite way, staying parallel to the axis
- **C.** It stays level, while the axis tilts away from it

**37.** A rotor head is built with no flapping hinges at all. How can dissymmetry of lift still be balanced on this rotor?

- **A.** The lead-lag hinges let the blades move up and down instead
- **B.** The blades themselves flex and absorb the forces acting on them
- **C.** The feathering hinges let the blades change pitch instead

**38.** Which turning force does a counterclockwise turning main rotor generate on the fuselage

- **A.** A nose-down pitching force
- **B.** A clockwise turning force
- **C.** A counterclockwise turning force

## EQUIPMENT AND FURNISHINGS

**39.** A microphone has been left out of its stowage and unused at a cabin crew station. What must the public address system do in that condition?

- **A.** Remain fully operative in the cabin
- **B.** Become inoperative until that microphone is stowed
- **C.** Mute the loudspeakers in that cabin zone

**40.** Which three things does the Cabin Intercommunication Data System do to the cabin systems it serves

- **A.** Controls, tests and monitors them
- **B.** Controls, records and displays them
- **C.** Tests, records and announces them

# Answer key

Answer, explanation and the manual page each question comes from

1. **A** Exhaust gas temperature indicators are named as engine monitoring devices under aircraft monitoring. (L2, PDF p. 347)
2. **B** The PFD is the screen that holds all the air data information, and it shows every one of the Basic T parameters. (L3, PDF p. 373)
3. **A** Of the three instruments only the airspeed indicator uses pitot pressure, so only it depends on the pitot probe. (L3, PDF p. 403)
4. **C** Static source error depends first of all on Mach number, and that value is supplied by the Mach module. (L3, PDF p. 499)
5. **B** The Mach module takes its information from the altitude and airspeed modules, both of which use the static pressure, so an error in that pressure carries into the Mach number. (L3, PDF p. 499)
6. **C** The falling speed of sound is the stated reason. (L2, PDF p. 477)
7. **C** For small height changes the fine value alone is used and the HIGH LEVEL detector blocks the coarse value. (L2, PDF p. 503)
8. **C** The step from EAS to TAS is the substitution of real air density for standard density. (L3, PDF p. 509)
9. **C** ARINC 565 is the analogue format, ARINC 429 the digital one. (L1, PDF p. 513)
10. **A** Those inputs are used for the data corrections, so only the correction accuracy suffers. (L3, PDF p. 515)
11. **B** Conversion to a digital data word happens at the module, right by the sensing point. (L2, PDF p. 539)
12. **C** Only the monitoring system records condition data that lets a part be used to its limit and still be changed at base. (L3, PDF p. 583)
13. **C** Storage beyond 25 hours comes from matching recorder capacity with data rate. (L3, PDF p. 591)
14. **B** In a glass cockpit the rate of turn appears on a display. (L2, PDF p. 695)
15. **A** The magnetic needle principle is the mechanical method and gives the direct reading compass. (L3, PDF p. 733)
16. **B** The minimum fit is two computers, so the third one carried for redundancy on large aircraft is missing. (L3, PDF p. 799)
17. **A** The ILS mode of the ND indicates localizer and glideslope deviation. (L1, PDF p. 849)
18. **A** Door closed is one of the items checked, so an open door prevents the normal result. (L2, PDF p. 867)
19. **C** They stay active until a positive rate of climb exists. (L2, PDF p. 913)
20. **B** The cockpit indication is normally given in units. (L1, PDF p. 947)
21. **B** The rules call for the modes to be annunciated to the pilot in colour. (L2, PDF p. 86)
22. **C** Engagement is the closing of the clutch between the AP-servo and the primary flight controls. (L1, PDF p. 96)
23. **A** The force becomes a proportional roll rate, and a lasting roll rate changes the attitude and so the heading. (L3, PDF p. 126)
24. **C** Where vertical speed is the basic mode, switching off any upper vertical mode leaves the autopilot in vertical speed. (L3, PDF p. 156)
25. **C** Above the transition level flight levels are flown on standard setting, so the target is a pressure altitude. (L3, PDF p. 166)
26. **B** The acquisition must already be running, not merely armed, for the hold to follow automatically. (L2, PDF p. 172)
27. **C** Both a heading error and a lateral displacement have to be removed, so both signals are used. (L3, PDF p. 224)
28. **A** Flying at the potential angle means the energy is exactly balanced, so both errors vanish. (L3, PDF p. 230)
29. **A** Inverse proportionality means the signal is reduced as speed increases. (L3, PDF p. 268)
30. **B** On failure of a fail passive system the pilot has to take over control. (L2, PDF p. 292)
31. **A** The hybrid system's reserve is manual flight on the supporting guidance system, not a second automatic system. (L3, PDF p. 300)

32. **C** A sideward input leaves the front servo unenergized and moves the helicopter sideward to the left. (L3, PDF p. 314)
33. **B** The attenuation signals go to the servo valves of the control servos. (L2, PDF p. 320)
34. **B** Of the two forces the airfoil produces, lift is the one the manual gives as counteracting gravity. (L2, PDF p. 22)
35. **A** The advancing blade moves forward, and the relative wind is always opposite to the blade's flight path. (L3, PDF p. 12)
36. **A** The plane of rotation is always perpendicular to the axis of rotation, so it tilts with the axis. (L3, PDF p. 16)
37. **B** Where there is no flapping hinge, the blades must instead have properties that allow them to absorb the forces. (L3, PDF p. 52)
38. **B** The fuselage force is opposite, so clockwise. (L1, PDF p. 56)
39. **A** No unused, un-stowed microphone may render the system inoperative, so the system has to remain fully operative. (L3, PDF p. 332)
40. **A** Control, test and monitor are the three functions printed for the CIDS. (L2, PDF p. 334)